

Mergeability Has a History

Training provenance determines removable representation mismatch

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Merging fails for two reasons, and the tooling cannot tell them apart

- ▶ Two models can be incompatible because they organise the same computation under different coordinates — or because they learned different things.
- ▶ The first is removable by a function-preserving map. The second is not.
- ▶ Which one you face decides whether alignment is worth running at all.

The question

When can representational alignment rescue model merging, and when can it not?

Design: fix architecture, parameter count, tokenizer, data budget, optimiser and evaluation. Vary *one factor of the training history* at a time — initialisation, data order, curriculum, first language — then apply alignment as an **intervention** and measure what it recovers.

The answer is narrower than the framing we began with

Alignment rescues merging when, and essentially only when, two models were initialised apart.

- ▶ Not how *different* two training histories were — only whether they started from the same weights.
- ▶ Data order, curriculum and even a different first language leave models in a shared coordinate system.
- ▶ The obstruction is fixed at step 0: training neither creates nor removes it.

Result 1: the largest composition costs are the most removable

what differs	n	naive	aligned	coord. share
initialisation	3	2.298	0.150	+ 93%
data order	4	0.553	0.705	−27%
curriculum, shared init	2	0.796	0.592	+26%
curriculum, new init	2	2.287	0.593	+ 74%
first language	2	1.412	0.524	+63%

Stitch penalty in nats/token. Initialisation costs about four times what data order costs — and the two costs are of *different kinds*: a fitted map removes 93% of the first and makes the second **worse** by 27%. That negative rung matters most: it is a case where alignment *must not* help, and it does not.

Result 2: 2,320 merges, cut on initialisation rather than on our own labels

	pairs	rescue	95% CI	improvement
indep. init (seed regime)	64	+0.244	[+0.163, +0.321]	73
indep. init (curric. reg.)	8	+0.238	[+0.214, +0.261]	94
shared init	208	-0.000	[-0.000, +0.000]	7

Two different experiments with different curricula give +0.244 and +0.238, intervals overlapping almost entirely, while 208 shared-initialisation pairs give -0.000. Bootstrapped over *pairs*, not rows. The diagnostic even corrected our own experimental metadata: two curriculum pairs were independently initialised, the weights said so with no access to training logs, and the functional outcome followed the diagnostic rather than the label.

Result 3: it is the symmetry, not the distance and not the flexibility

Distance-matched

Interpolating B toward A by the same distance the map removes buys +0.036; the map buys +**1.405**.

39×

Divergence-matched

A pair placed equally far apart with its removable component exhausted by construction recovers **exactly** 0.000 — in all 18 cells.

Which symmetry class?

Stitch penalty, deep seams:

	identity	best
permutation S_1	1.880	0.474
orthogonal S_3	1.880	0.132
$GL(d)$ S_4	1.880	2.128

$GL(d)$ contains every other class and fits its own objective best — and it is **worse than doing nothing**. It does not commute with RMSNorm, so it is not a symmetry at all.

Alignment works through symmetry, not distance reduction.
Symmetry beats flexibility.

Result 4: is this true by construction? No — and we measured it

The objection: if two runs start from identical weights and no permutation is applied, units keep their identity *by definition*.

The testable alternative:
representational drift — units gradually change role over a long run, so the best correspondence stops being the identity.

checkpoint	shared init	indep. init
step-0	1.0000	0.0003
step-128	1.0000	0.0003
step-2048	1.0000	0.0003
main	1.0000	0.0003

Chance is $1/768 = 0.0013$.

Drift does not happen, at all

The minimum over *every layer of every checkpoint* is also 1.0000: not one unit out of 3072×14 ever swaps. The claim is not definitional — drift was a live alternative and is measured to be exactly zero — and the obstruction is therefore **fixed at step 0**.

Result 5: it holds outside our zoo, beyond two models, and across languages

Another lab's models

PolyPythia, 14M / 70M / 160M: the residual-basis factor survives acceptance on **74 of 74** independently initialised pairs. The detector never misses. Rescue falls with scale (+23.6, +6.1, +1.5).

Three language pairs

Coordinate share +92% (nld), +89% (deu), +93% (zho), against -26% for the shared-init rung. Same dissociation, including a non-Indo-European pair.

More than two models

<i>k</i>	2	3	4	
naive	-1.86	-3.28	-3.74	-3.9
aligned	-0.99	-1.33	-1.38	-1.4
rescue	+0.87	+1.95	+2.36	+2.4

Naive merging collapses as models are added; the aligned merge barely moves. Alignment's value grows with merge size — the strongest practical statement in the paper.

Takeaways

1. Alignment does not make models mergeable. It removes *one* obstruction, present when and essentially only when two models were initialised apart.
2. Provenance is the index, not similarity. Data order, curriculum and first language create large functional divergence with *no* coordinate component.
3. Group membership is the currency. Not distance, not flexibility — the richest map, $GL(d)$, is worse than no map at all.
4. The obstruction is fixed at step 0 and never decays. Cost of composing grows $916\times$ over training while the removable share stays above 92%.
5. Decidable in advance. Held-out AUROC 0.944 from the weights alone — though a plain weight cosine reaches 0.933, because the underlying fact is binary.

Honest limit: these are $\leq 200M$ models pretrained from scratch, where merging is destructive in absolute terms (-1.06 naive, -0.82 aligned). Alignment moves

Whatever pretraining does to make two models incompatible,
it is not making their difference irreducible.

Full deck, every figure and table: `main.pdf`